

Negative Literature Answer to the Classical Schreier Uniform λ -Property Question

FA/Banach run status packet

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Status

`literature_already_answered` (classical Schreier subquestion). This note records an exact answer by a separate later paper; it is not an original result of the run.

Original question

Antunes, Beanland, and Chu ask on page 9 of *On the geometry of higher order Schreier spaces* [1], immediately before Theorem 3.1, whether the classical Schreier space X_{S_1} has the uniform λ -property. Their same paragraph separately asks about the ordinary λ -property of $X_{\mathcal{F}}$ for every regular family $\mathcal{F}$.

Supporting answer

Beanland and Chu explicitly state that their first main theorem solves the problem restated in [1].

Literature Answer (Beanland–Chu, Theorem 1.1). *The classical Schreier space X_{S_1} does not have the uniform λ -property.*

In Section 3 of [2], they construct, for sufficiently large n , unit vectors x_n and show that every extremal decomposition

$$x_n = \lambda e + (1 - \lambda)y, \quad e \in E(X_{S_1}), \quad y \in Ba(X_{S_1}),$$

satisfies

$$\lambda \leq \frac{n+1}{n^2}.$$

The upper bound tends to zero. Hence no positive constant can uniformly lower-bound the λ -function on the unit ball. Theorem 1.2 of the same paper also gives the negative answer for the dual space.

Scope limitations

This answer clears only the classical X_{S_1} uniform-property question. It does not answer the broader regular-family question in the same source paragraph. The supporting paper lists uniformity for higher-order Schreier spaces and their duals as future research and conjectures a negative answer.

Verification

The source question is on page 9 of arXiv:1903.03492. The answer is Theorem 1.1 on page 1 of arXiv:1904.10384, with proof on pages 4–6. The answer paper names the earlier paper and says that Theorem 1.1 solves the problem restated there. The sources are distinct and the answer paper was posted later. An exact-id and keyword search found no pre-existing run packet for this question.

References

- [1] L. Antunes, K. Beanland, and H. V. Chu, *On the geometry of higher order Schreier spaces*, arXiv:1903.03492.
- [2] K. Beanland and H. V. Chu, *The Schreier space does not have the uniform λ -property*, Proc. Amer. Math. Soc. 149 (2021), 5131–5137; arXiv:1904.10384; DOI: 10.1090/proc/14766.